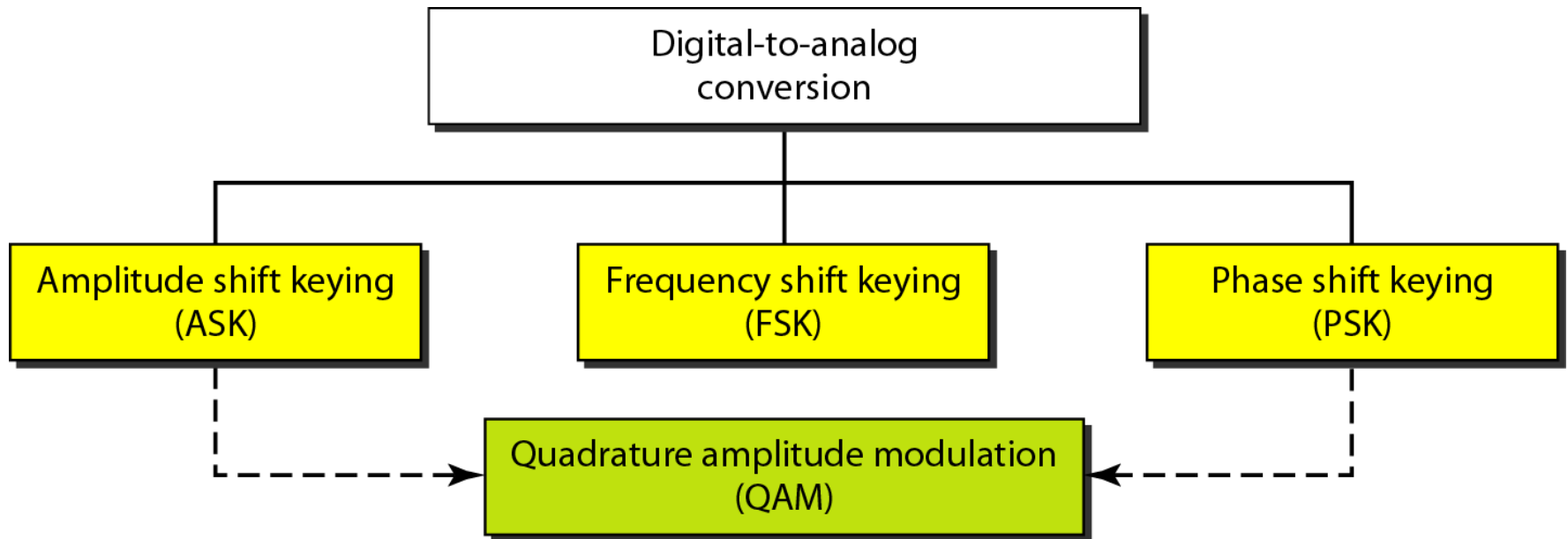
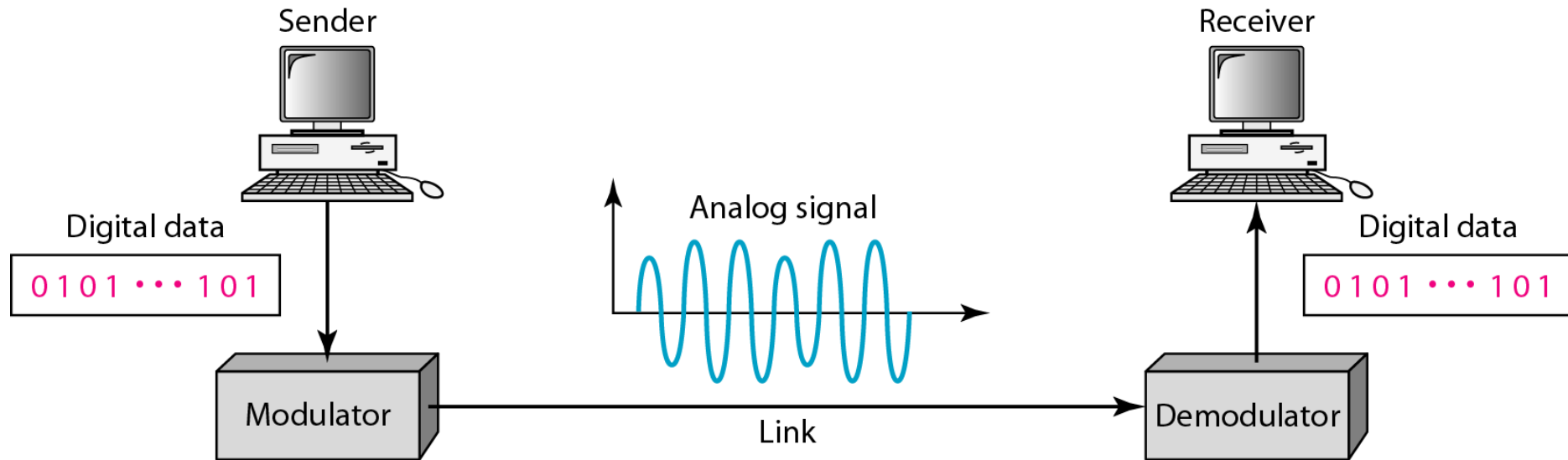


Types of digital-to-analog conversion



Digital-to-analog conversion



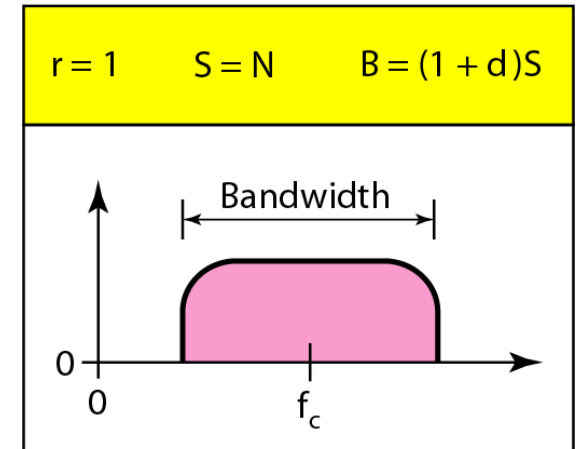
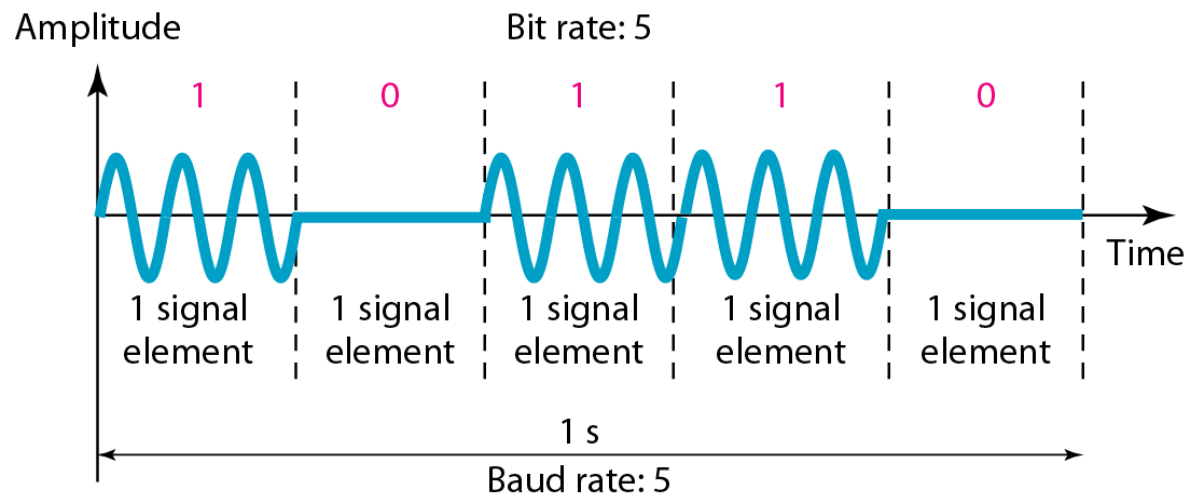
Note

**Bit rate, N , is the number of bits per second (bps). Baud rate is the number of signal elements per second (bauds).
In the analog transmission of digital data, the signal or baud rate is less than or equal to the bit rate.**

Amplitude Shift Keying (ASK)

- ASK is implemented by changing the amplitude of a carrier signal to reflect amplitude levels in the digital signal.
- For example: a digital “1” could not affect the signal, whereas a digital “0” would, by making it zero.
- The line encoding will determine the values of the analog waveform to reflect the digital data being carried.

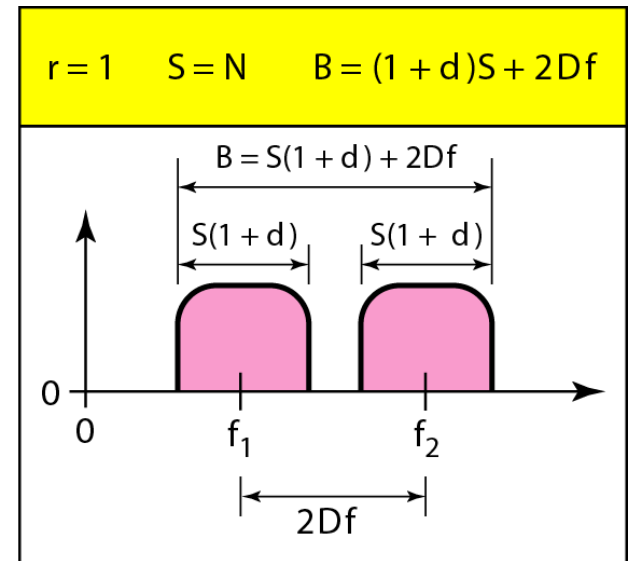
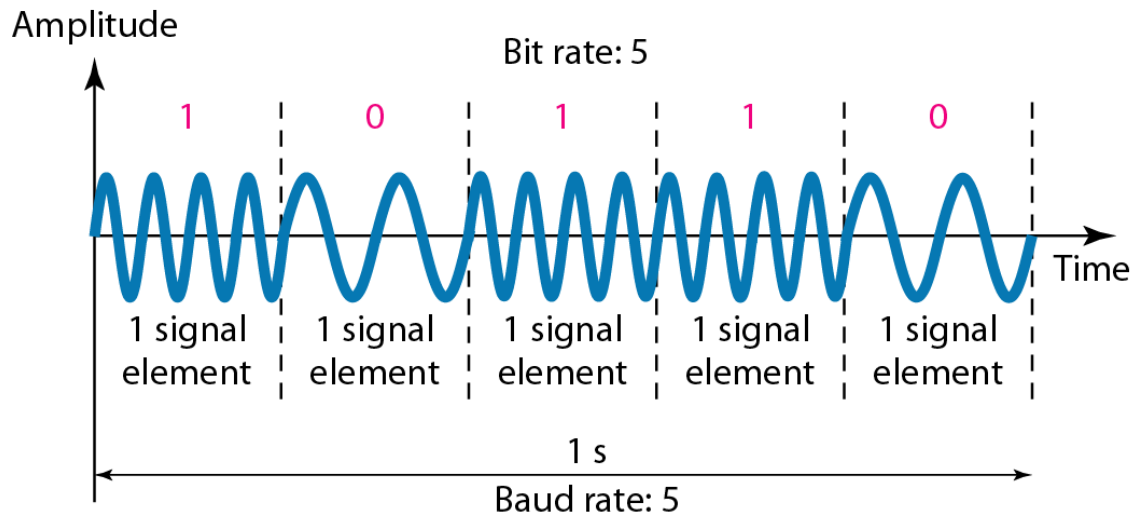
Figure 5.3 *Binary amplitude shift keying*



Frequency Shift Keying

- The digital data stream changes the frequency of the carrier signal, f_c .
- For example, a "1" could be represented by $f_1 = f_c + \Delta f$, and a "0" could be represented by $f_2 = f_c - \Delta f$.

Figure 5.6 *Binary frequency shift keying*



Binary phase shift keying

